ECO102: Introduction to Macroeconomics

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SECTION: 06

LECTURE 16

CHAPTER 24: MONEY, THE PRICE LEVEL AND, INFLATION (MONEY MARKET)

Central Bank (read from the book)

- The Central bank influences the quantity of money and interest rates by adjusting the quantity of reserves available to the banks and the reserves the banks must hold. To do this, the central bank manipulates three tools:

- Open market operations: Open market purchase and sell.
- Last resort loans
- Required reserve ratio

The Money Market

The quantity of money that people plan to hold depends on four main factors:

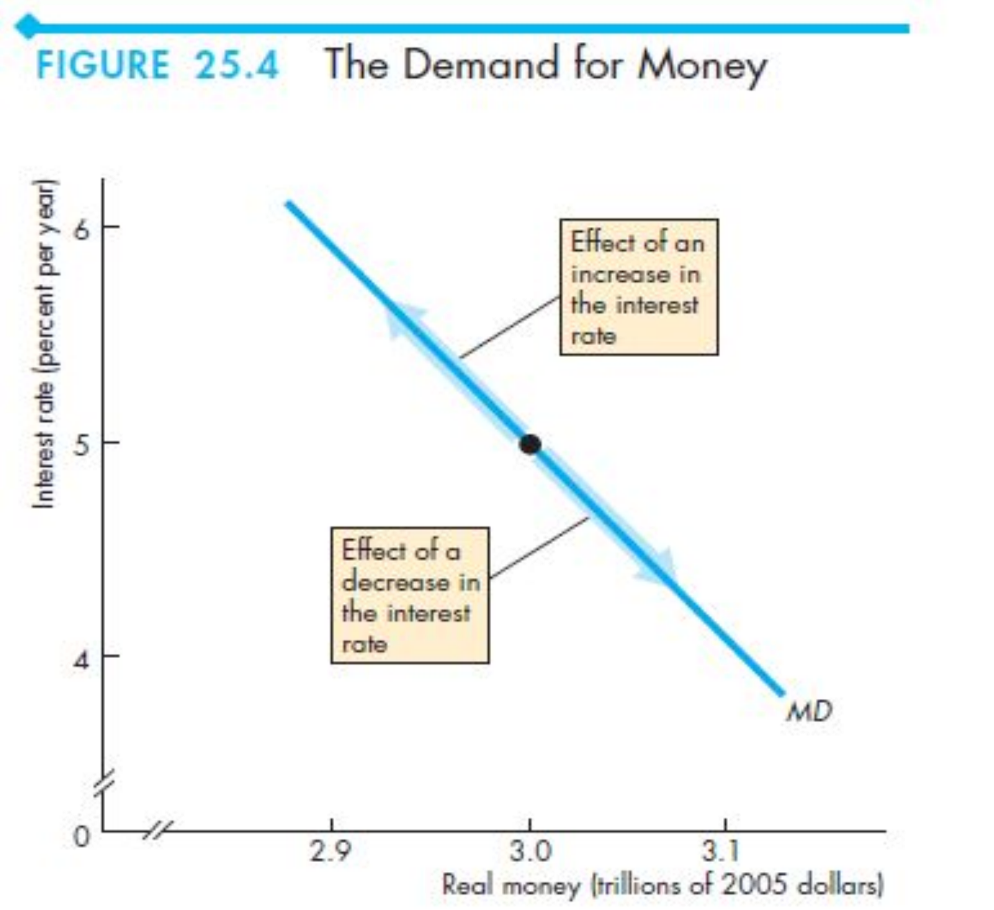
- The price level
- The *nominal* interest rate
- Real GDP
- Financial innovation

The Demand for Money

- The **demand for money** is the relationship between the quantity of real money demanded and the nominal interest rate when all other influences on the amount of money that people wish to hold remain the same.

- When any influence on money holding other than the interest rate changes, there is a change in the demand for money and the demand for money curve shifts.

- For only change in interest rate there is a movement along the curve.

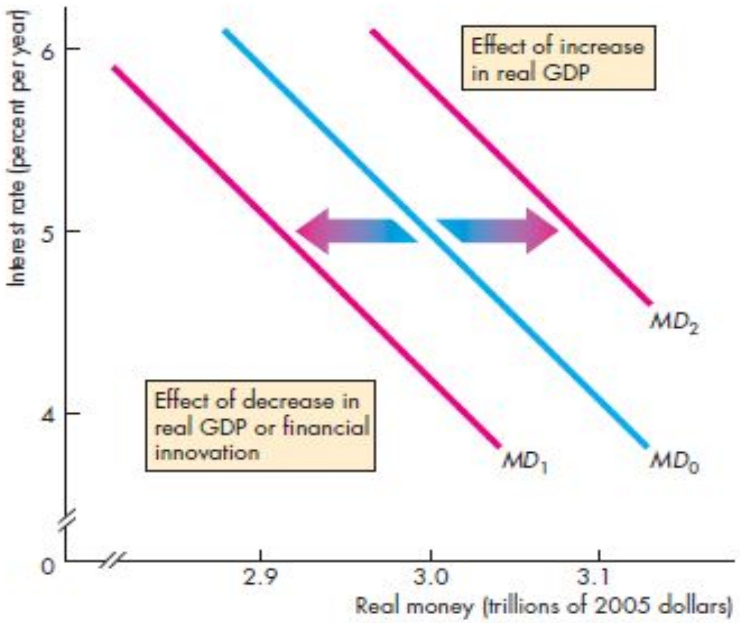


Changes in the Money Demand

- Figure 25.5 illustrates the change in the demand for money. A decrease in real GDP decreases the demand for money and shifts the demand for money curve leftward from *MD*0 to *MD*1.

- An increase in real GDP has the opposite effect: It increases the demand for money and shifts the demand for money curve rightward from *MD*0 to *MD*2.

FIGURE 25.5 Changes in the Demand for Money

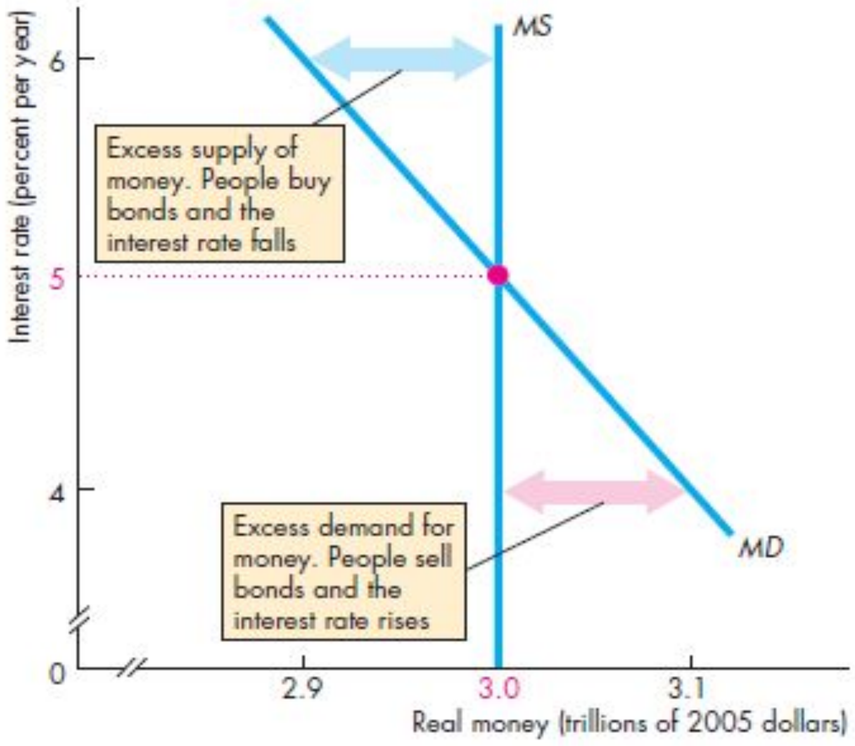


A decrease in real GDP decreases the demand for money. The demand for money curve shifts leftward from MD_0 to MD_1. An increase in real GDP increases the demand for money. The demand for money curve shifts rightward from MD_0 to MD_2. Financial innovation generally decreases the demand for money.

Money Market Equilibrium (short-run)

- Money market equilibrium occurs when the quantity of money demanded equals the quantity of money supplied.
- The quantity of money supplied is determined by the actions of the banks and the central bank.
- As the central bank adjusts the quantity of money, the interest rate changes.
- If interest rate was less people would want to hold more money than available
- If interest rate was more people would want to hold less money that available.

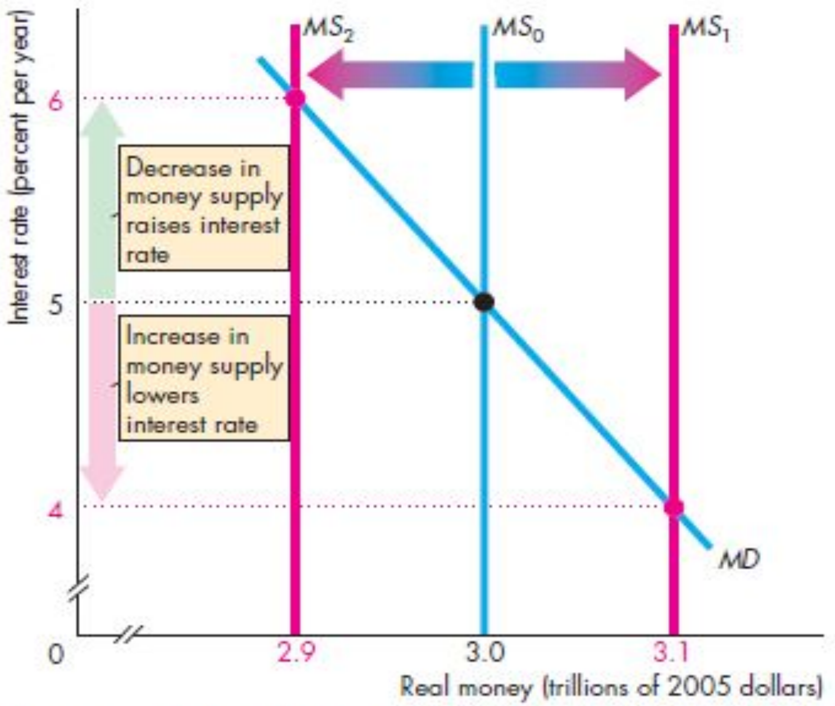
FIGURE 25.6 Money Market Equilibrium



A Change in the Supply of Money

- If CB increases the quantity of money, people find themselves holding more money than the quantity demanded-
- With a surplus of money holding, people enter the loanable funds market and buy bonds. The increase in demand for bonds raises the price of a bond and lowers the interest rate.
- Vice versa if the CB decreases the quantity of money people sell bonds in the loanable funds market and decreased demand for bonds raises the interest rate.

FIGURE 25.7 A Change in the Supply of Money



An increase in the supply of money shifts the supply of money curve from MS_0 to MS_1 and the interest rate falls. A decrease in the supply of money shifts the supply of money curve from MS_0 to MS_2 and the interest rate rises.

The Long Run Equilibrium

- When the inflation rate equals the expected (or forecasted) inflation rate and when real GDP equals potential GDP, the money market, the loanable funds market, the goods market, and the labor market are in long-run equilibrium—the economy is in long run equilibrium.
- If in long-run equilibrium nothing 'real' changes, only the price level changes. Why?

Ans: The reason is that

-real GDP and employment are determined by the demand for labor, the supply of labor, and the production function; and

-the real interest rate is determined by the demand for and supply of (real) loanable funds.

The only variable that is free to respond to a change in the supply of money in the long run is the price level. The price level adjusts to make the quantity of real money supplied equal to the quantity demanded.

The Transition from Short run to Long run

- We start out in long-run equilibrium and the CB increases the quantity of money.
- Nominal interest rate fall, and real interest rate falls too as people try to get rid of excess money.
- With lower interest rate people want to borrow and spend more.
- The increase in demand for goods cannot be met by an increase in supply because the economy is already in full employment.
- So, there is a shortage of goods and services which forces the price level to rise.
- As the price level rises, the real quantity of money decreases.
- The decrease in the quantity of real money raises the nominal interest rate and the real interest rate.
- As the interest rate rises, spending plans are cut back, and eventually the original full-employment equilibrium is restored.
- At the new long-run equilibrium, the price level has risen by the same percent and nothing real has changed.

The Quantity Theory of Money (optional)

- The **quantity theory of money** is the proposition that in the long run, an increase in the quantity of money brings an equal percentage increase in the price level.

- The **velocity of circulation** is the average number of times a dollar of money is used annually to buy the goods and services that make up GDP
- $GDP = PY$, here, P= price level, Y= Real GDP
- $V = PY/M$, here V= velocity of circulation, M= quantity of money
- $MV = PY$
- $P = M(V/Y)$; here, (V/Y) is independent of M.
- Quantity of money does not influence the velocity of circulation or real GDP. So a change in M brings a proportional change in P.